

Data Flow Diagram (DFD)

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Project Name:	ANTI DEBT	Lead:	Lakshmi Sravya Kagitha (Team Lead)

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded	External Entity	Borrower submitting credentials or request inputs.
Rectangle (Header)	Process	Execution processes like Auth Handler, Ratios calculation, Gemini API caller.
Solid Line Box	Data Store	SQLite Database storing users, loan variables, recommendations, letters, and logs.
Arrow Line	Data Flow	Flow of data fields between borrower, system processors, and databases.

DFD Level 0 Diagram Summary

- Borrower sends profile fields and credential values -> Auth Handler.
- User posts loan variables -> DB storage.
- User requests AI settlement options -> Gemini processors compute and return target terms.
- System outputs dashboard charts, PDF files, and Excel tables back to Borrower.

DFD Level 1 Process Flow

- Process 1.0 (Authenticate User): Reads/writes SQLite storage to verify login tokens.
- Process 2.0 (Calculate Ratios): Queries DB for all active user loans, runs DTI and stress index formulas, then writes analysis records back.
- Process 3.0 (Query Gemini AI): Packages active DTI metrics and profile logs into prompts, sends request, parses JSON response structure, and commits recommendations.